

Name:

Unique ID:

Math 221D Quiz 2 (September 11, 2026)

- (a) (15 points) Find all vectors $\mathbf{x} \in \mathbb{R}^5$ which are orthogonal to $(1, 0, 2, 4, 3)$, $(2, -1, 1, 7, 5)$, and $(-1, 3, 7, 5, 6)$. Express your answer in the form $\text{Span}(\mathbf{v}_1, \mathbf{v}_2)$ for some vectors $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^5$.

Part (b) on the back of this page!

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- (b) (5 points) Conceivably, your classmates could give a different pair of $\mathbf{v}_1$ and $\mathbf{v}_2$ for part (a) and still be correct. However, show that it is not possible to express your answer to part (a) in the form $\text{Span}(\mathbf{v})$ for a *single* vector $\mathbf{v} \in \mathbb{R}^5$.